Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

Title: Connecting Power BI Desktop to Flat Files (Excel/CSV) and MySQL Database for Retail Sales Analysis"

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Experiment No.: 1

Importing Business Datasets from Excel/CSV & MySQL and Connecting with Other Sources into Power BI

Problem Statement

In this experiment, I worked as a Data Analyst for a retail company where the business data was spread across different sources - sales records in an Excel file, customer details in a CSV file, and product information in a MySQL database. Since management wanted one centralized place to view all this data for reporting, the goal was to bring all three sources into Power BI, connect them properly, and prepare them for analysis and dashboard creation.

Methodology / Steps Followed

Step 1: Data Collection

- Collected the Sales Data (Order ID, Product Category, Customer ID, Date, Quantity, Price per Unit, Total Amount) as an Excel file.
- Collected the Customer/Store Data (Customer ID, Customer Name, City Type, Region, Segment) as a CSV file.
- Collected the Product Data (ProductID, ProductName, Price) and stored it in a MySQL database (retail_db.products table), since this was meant to represent data coming from a separate database source.

Step 2: Data Import

- Opened Power BI Desktop and started a new report.
- Used Get Data → Excel Workbook to bring in the Sales dataset.
- Used Get Data → Text/CSV to bring in the Customer dataset.
- Set up a MySQL database named retail_db and created a products table in MySQL Workbench, then imported the product records into it using the Table Data Import Wizard.
- Connected Power BI to this MySQL database using Get Data → Database → MySQL database, and loaded the products table into the report.

Step 3: Data Cleaning and Transformation

- Checked that each column had the correct data type (for example, dates as Date and amounts as numbers) after loading.
- Looked for duplicate customer and product entries and removed them where found.
- Renamed a few columns so that they were easier to understand and consistent across all three tables.
- Used Power Query to handle any missing or blank values in the data.

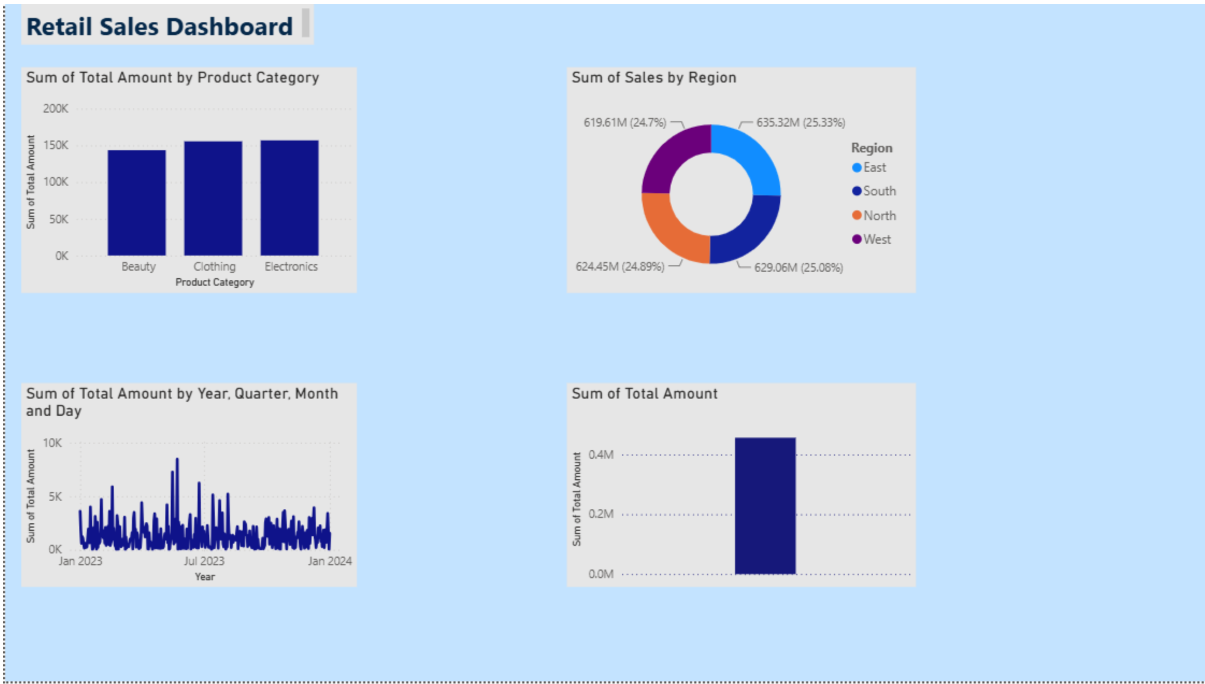
Step 4: Data Modeling

- Switched to Model view in Power BI to see all three tables together.
- Built a relationship between the Sales table and the Customer table using the common Customer ID column (one-to-one).
- Built a relationship between the Sales table and the Products table using Product Category linked to Product Name (many-to-one).
- Opened Manage Relationships to confirm both relationships were set as Active.

Step 5: Dashboard Development

- Built a bar chart showing Total Amount by Product Category to analyze Sales by Product.
- Built a bar chart showing Total Sales by Region to analyze Sales by Store/Region.
- Built a line chart showing Total Amount over Year, Quarter, Month and Day to analyze Sales by Time.
- Added a visual summarizing the overall Total Amount so the total sales figure is visible at a glance.
- Applied a consistent color theme across all visuals and added a proper dashboard title to give the report a clean, professional look.

Output



Once completed, the Power BI dashboard was able to show the following:

- Overall Total Sales Performance
- Product-wise Sales Analysis

- Regional Sales Performance
- Monthly and Yearly Sales Trends
- One combined data model built from Excel, CSV and MySQL sources

With this setup, management can look at business performance directly through the dashboard instead of checking the Excel file, the CSV file, and the database separately each time.

Conclusion

Through this experiment, I was able to bring together data from three different sources - Excel, CSV, and MySQL - into a single Power BI report. Along the way, I cleaned and transformed the data, built relationships between the tables, and used that connected data model to create a working dashboard. The final result gives a combined view of sales across products, regions, and time, which is much easier for decision-making than looking at three separate files.

Learning Outcomes

From this project, I was able to learn:

- How to import data into Power BI from both Excel and CSV files.
- How to set up and connect Power BI to a MySQL database and pull in tables from it.
- How to clean and transform data using Power Query before analysis.
- How to create and manage relationships between tables coming from different sources.
- How to design a dashboard with multiple visuals that work together.
- How to read raw retail data and turn it into insights that are actually useful for a business.